

Vague Preferences and Contracts

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Which of the following are green:



(a)



(b)



(c)

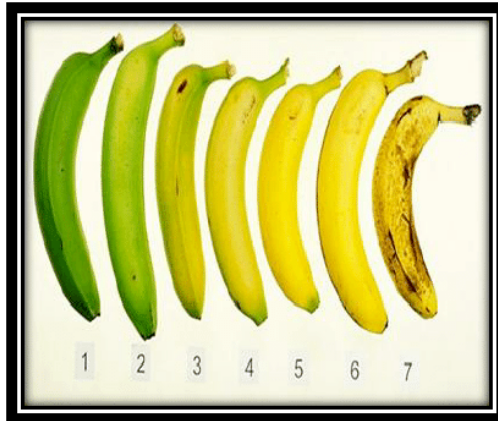


(d)



(e)

Which of the following are ripe:



Vagueness

The language we use is less precise than the reality it describes.

- ◇ Descriptions that appear semantically crisp are in fact nebulous
- ◇ A statement that is neither (absolutely) true nor false is **vague**
- ◇ Contrasted with 'uncertainty': when the truth of statements is not known

Vagueness is economically relevant—a principal function of the (tort) legal system is to determine *the degree of truth*. Consider:

“Gilead Science’s Hepatitis-C treatment, sofosbuvir, infringes on Idenix Pharmaceutical’s patent.”

- ◇ Essentially: Idenix claimed that the structure of sofosbuvir is based on already known molecules
- ◇ The truth of which was eventually settled by a jury to the tune of 2.5 billion dollars

This paper...

- ◇ Constructs a framework for modeling vagueness
- ◇ Provides a methodology for eliciting decision maker's subjective beliefs
- ◇ Examines how vagueness affects strategic contracting (principal-agent style)

Language is Important

- ◇ Vagueness arises from the gap between descriptions of reality and reality itself
- ◇ It's not clear *what* a state-space should look like, and if it would be describable in the actual language
- ◇ So, begin with a formal language, $\mathcal{L}$

Language is Important

The language $\mathcal{L}$ represents payoff relevant parameters, constructed by

- ◇ $\mathcal{P}$ a set of atomic propositions
- ◇ For any $\varphi, \psi \in \mathcal{L}$ we have
 - ◇ $\neg\varphi$: not φ
 - ◇ $\varphi \rightarrow \psi$: φ implies ψ

Language is Important

It will be notationally convenient to define several other operations from implication and negation:

$$\diamond \quad \varphi \vee \psi =_{def} (\varphi \rightarrow \psi) \rightarrow \psi$$

$$\diamond \quad \varphi \wedge \psi =_{def} \neg(\neg\varphi \vee \neg\psi)$$

$$\diamond \quad \varphi \oplus \psi =_{def} \neg\varphi \rightarrow \psi$$

$$\diamond \quad \varphi \odot \psi =_{def} \neg(\varphi \rightarrow \neg\psi)$$

Valuations

To allow for partial-truth, vagueness, fuzziness, what-have-you, we value statements via $v : \mathcal{L} \rightarrow [0, 1]$

- ◇ $v(\varphi) = 0$ indicates φ is **absolutely false**
- ◇ $v(\varphi) = 1$ indicates φ is **absolutely true**
- ◇ $v(\varphi) > v(\psi)$ indicates φ is **more true** than ψ

Valuations

A function $v : \mathcal{L} \rightarrow [0, 1]$ is a valuation if

$$\llbracket \neg \rrbracket \quad v(\neg \varphi) = 1 - v(\varphi)$$

$$\llbracket \rightarrow \rrbracket \quad v(\varphi \rightarrow \psi) = \min\{1, 1 - v(\varphi) + v(\psi)\}$$

which implies:

$$\llbracket \vee \rrbracket \quad v(\varphi \vee \psi) = \max\{v(\varphi), v(\psi)\}$$

$$\llbracket \wedge \rrbracket \quad v(\varphi \wedge \psi) = \min\{v(\varphi), v(\psi)\}$$

$$\llbracket \oplus \rrbracket \quad v(\varphi \oplus \psi) = \min\{1, v(\varphi) + v(\psi)\}$$

$$\llbracket \odot \rrbracket \quad v(\varphi \odot \psi) = \max\{0, v(\varphi) + v(\psi) - 1\}$$

Valuations

- ◇ If we interpret “or” as:
 - ◇ $\vee$: maximum — “Man is evil or man is not evil”
 - ◇ $\oplus$: (truncated) summation — “The rectangle is green or its yellow”
- ◇ If v sends $\mathcal{P}$ to $\{0, 1\}$ then all statements are $\{0, 1\}$ -valued; this is classic logic
- ◇ For classical logic $\vee$ and $\oplus$ coincide
- ◇ $\wedge$ and $\odot$ are dual

Decision Theory

- ◇ An **act** is a function $f: \mathcal{L} \rightarrow \mathbb{R}_+$, finite support.
- ◇ A **bet** x_φ is the act that maps φ to x and all other statements to 0.
- ◇ Bets are the extreme points of the set of acts
- ◇ The primitive is $\succsim$ over acts.

Decision Theory

Interpretation: Payoffs depend on truth values but contracts can only be written using the language $\mathcal{L}$

- ◇ A bet x_φ is less valuable the less true x is

E.g., x_φ is an investment in a project, φ is the statement that the project does not infringe on intellectual property

Standard Axioms

A1 **Order** $\succsim$ is a non-trivial, continuous weak order.

A2 **Payoff Monotonicity** if f point-wise dominates g then $f \succsim g$.

A3 **Independence** $f \succsim g$ if and only if $\alpha f + (1 - \alpha)h \succsim \alpha g + (1 - \alpha)h$.

Axiom: Łukasiewicz Consistency

Call φ and ψ **disjoint** if $v(\varphi \odot \psi) = 0$ for any valuation v :

- ◇ Disjointness is tantamount to $\varphi \rightarrow \neg\psi$ (and vice-versa)
 - ◇ In classical logic, φ and ψ can never be true at the same time
 - ◇ Allowing for vagueness: the more true φ is, the less true ψ must be

Axiom: Łukasiewicz Consistency

A4 **Łukasiewicz Consistency** If φ and ψ are disjoint then:

$$\frac{1}{2}\varphi + \frac{1}{2}\psi \sim \frac{1}{2}\varphi \oplus \psi,$$

Theorem

$\succsim$ satisfies A1-4 if and only if it is there exists a *vague model*, (Ω, V, μ) ,

- ◇ Ω is a topological state space,
- ◇ $V = \{v_\omega : \mathcal{L} \rightarrow [0, 1]\}_{\omega \in \Omega}$ is Borel measurable
- ◇ μ a regular Borel probability measure over Ω .

such that

$$f \succsim g \iff \int_{\Omega} f^V \, \mathrm{d}\mu \geq \int_{\Omega} g^V \, \mathrm{d}\mu \quad (\star)$$

where $f^V : \Omega \rightarrow [0, 1]$ is defined as

$$f^V : \omega \mapsto \sum_{\varphi \in \mathcal{L}} f(\varphi) v_\omega(\varphi).$$

Moreover this model is unique up-to isomorphism.

Representation

Given a model (Ω, V, μ) and an act f ,

$$f^V : \omega \mapsto \sum_{\varphi \in \mathcal{L}} f(\varphi) v_{\omega}(\varphi).$$

yields the ‘weighted’ payoff of f .

If $f = x_{\varphi}$, then $f^V(\omega) = x v_{\omega}(\varphi) = (\text{payoff}) \times (\text{truth value})$

Representation

The the value for a bet x_φ is:

$$\int_{\Omega} x - v_{\omega}(\varphi) \, \mathrm{d}\mu$$

Representation

The the value for a bet x_φ is:

vagueness 

$$\int_{\Omega} x \quad v_{\omega}(\varphi) \quad \mathrm{d}\mu$$

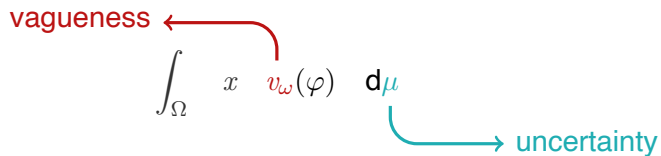
Representation

The the value for a bet x_φ is:

vagueness $\leftarrow$

$$\int_{\Omega} x \quad v_{\omega}(\varphi) \quad d\mu$$

$\rightarrow$ uncertainty



Representation

This model assumes linearity in both probability (expected utility) and in truth value:

- ◇ This is necessary if we want $x_{\varphi} + x_{\neg\varphi} = x_{TRUE}$
- ◇ This is unfortunate

Contracting — Principal Agent

Principal Agent Model

- ◇ Let Ω denote a state-space, the states of which are associated with the various outcomes of the project.
- ◇ Agent chooses unobservable $e \in E$;
 - ◇ agent pays a utility cost $c(e) \in \mathbb{R}$,
 - ◇ induces μ_e , distribution over Ω
- ◇ The agent's continuously differentiable and strictly monotone utility index over money is $u : \mathbb{R} \rightarrow \mathbb{R}$: her ex-post utility is $u(x) - c(e)$.
- ◇ Outside option is $\bar{u} \in \mathbb{R}$.

Principal Agent Model

Departure from the standard model: Ω is not directly contactable.

- ◇ There exists a language, $\mathcal{L}$ and a valuation $V = \{v_\omega\}_{\omega \in \Omega}$
- ◇ The Principal must write an actual (linguistic) contract
- ◇ Each contract induces a function $f: \Omega \rightarrow \mathbb{R}$,
- ◇ Not all such functions might be induceable — each ‘contract writing technology’ is associated with $C \subseteq \mathbb{R}^\Omega$

Principal Agent Model

Say that a contract $f \in \mathbb{R}^\Omega$ *implements* $e \in E$ if

$$e = \arg \max_{e' \in E} \int_{\Omega} u \circ f \, \mathrm{d}\mu_{e'} - c(e') \quad (\text{IC})$$

$$\int_{\Omega} u \circ f \, \mathrm{d}\mu_e \geq \bar{u} \quad (\text{IR})$$

Let $\mathcal{L}$ be constructed from $\mathcal{P} = \mathbf{p}_1 \dots \mathbf{p}_n$, and consider the set of contracts:

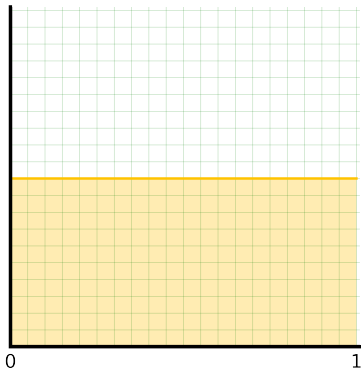
$$C^{\mathcal{P}} = \{xv_{\omega}(\mathbf{p}) + yv_{\omega}(\neg\mathbf{p}) \mid x, y \in \mathbb{R}, \mathbf{p} \in \mathcal{P}\}.$$

- ◇ $f \in C^{\mathcal{P}}$ is predicated directly on the truth of the propositions in $\mathcal{P}$.
- ◇ $xv_{\omega}(\mathbf{p}) + yv_{\omega}(\neg\mathbf{p})$ induces the affine function $f: \omega \mapsto (x - y)v_{\omega}(\mathbf{p}) + y$
- ◇ $C^{\mathcal{P}}$ is the set of all affine contracts

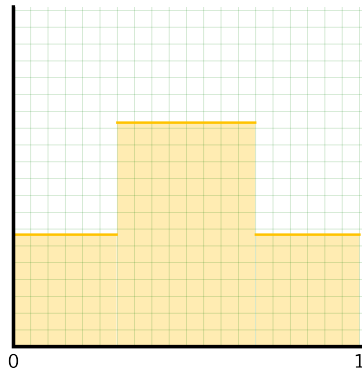
There are outcomes implementable by direct contracts (i.e., continuous functions over Ω) not implementable by C^P . This is obvious; for example:

- ◇ Single p , $\Omega = [0, 1]$ is the truth of p .
- ◇ $E = \{e, e'\}$:
 - ◇ e induces uniform measure, less costly
 - ◇ e' concentrates probability symmetrically around $\frac{1}{2}$, more costly
- ◇ u is linear

Distribution of Outcomes

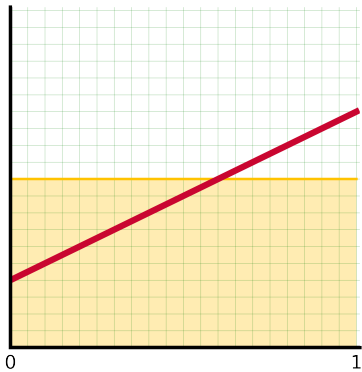


μ_e

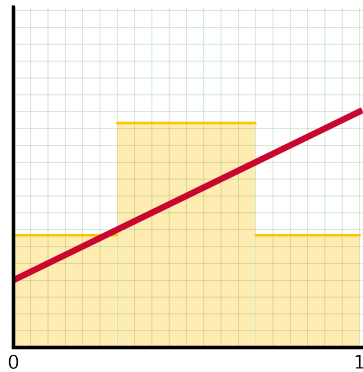


$\mu_{e'}$

Contracts in $C^{\mathcal{P}}$ cannot implement e'

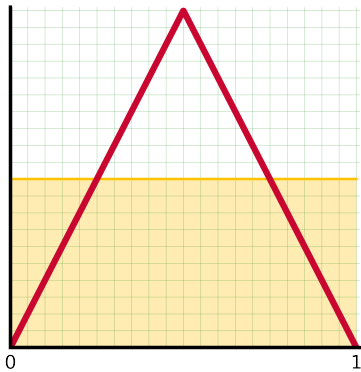


μ_e

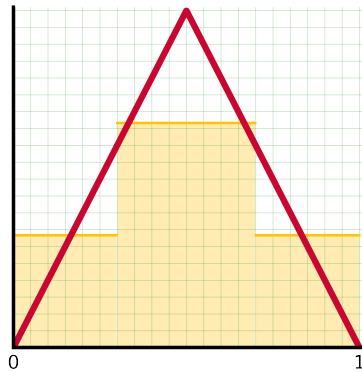


$\mu_{e'}$

Contracts not in $C^{\mathcal{P}}$ can implement e'



μ_e



$\mu_{e'}$

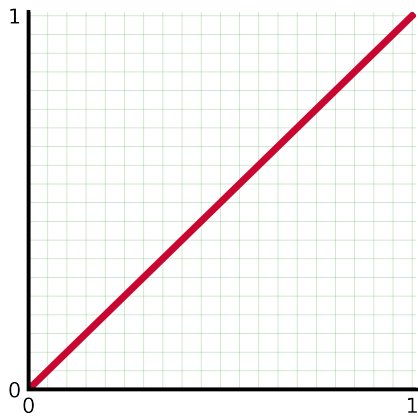
What about a richer set of linguistic contracts:

$$C^{\mathcal{L}} = \{xv_{\omega}(\varphi) + yv_{\omega}(\neg\varphi) \mid x, y \in \mathbb{R}, \varphi \in \mathcal{L}\}.$$

- ◇ $f \in C^{\mathcal{L}}$ is predicated directly on compound statements.
- ◇ Still constructable contracts, but over more complex language

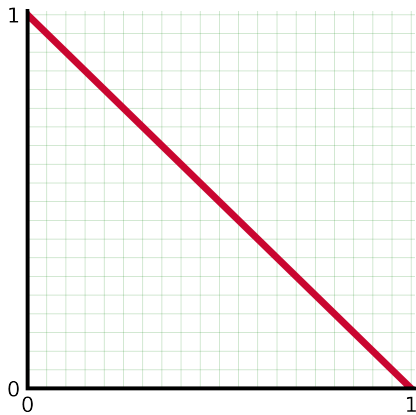
$$\mathbf{p} \mapsto 1$$

$$\omega \mapsto \omega$$



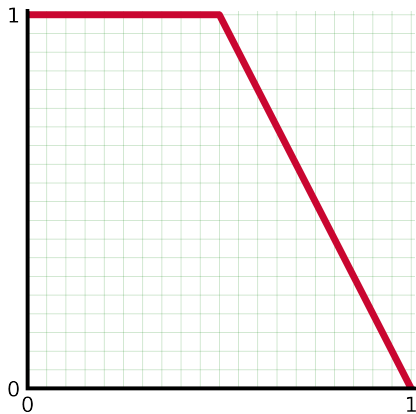
$$\neg \mathbf{p} \mapsto 1$$

$$\omega \mapsto 1 - \omega$$



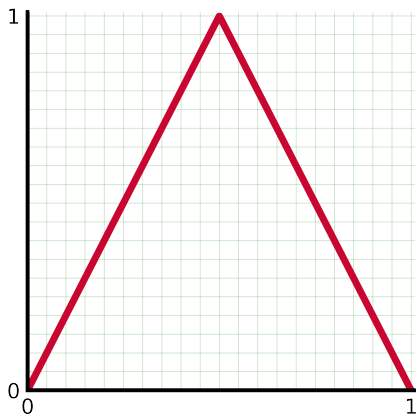
$$(\mathbf{p} \rightarrow \neg \mathbf{p}) \mapsto 1$$

$$\omega \mapsto \min\{1, 1 - \omega + (1 - \omega)\} = \min\{1, 2 - 2\omega\}.$$



$$(\mathbf{p} \rightarrow \neg \mathbf{p}) \wedge (\neg \mathbf{p} \rightarrow \mathbf{p}) \mapsto 1$$

$$\omega \mapsto \min \left\{ \min\{1, 1 - \omega + (1 - \omega)\}, \min\{1, 1 - (1 - \omega) + \omega\} \right\} \\ = \min\{1, 2 - 2\omega, 2\omega\}.$$



Theorem

$C^{\mathcal{L}}$ implements the same outcomes, at arbitrarily close cost, as the set of all direct contracts (continuous maps from Ω).

- ◇ Subject to some mild regularity conditions, of course
- ◇ Even though the Principal cannot directly condition on Ω , the calculus of $\mathcal{L}$ is rich enough to approximate

Thank You